

Raghumani Mehta

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Education

University of British Columbia

B.Sc. Computer Science, Minor in Physics

Dean's List (2024–2025)

Vancouver, BC

Expected Graduation: May 2028

Technical Skills

- **Languages:** Python, C++, C, Rust, Java, TypeScript, JavaScript, SQL
- **Frameworks & Tools:** Node.js, Express, Spring Boot, Git, Docker, ROS, GitHub Actions
- **Systems:** Linux/Unix, POSIX, Sockets
- **Cloud:** AWS (Textract, Lambda, S3, SQS, SNS, RDS)
- **Testing:** JUnit, Mocha/Chai, Pytest, GDB, Valgrind

Technical Work Experience

Software Co-Lead | github.com/UBCSailbot/sailbot_workspace

Feb 2025 – Present

UBC Sailbot (Python, ROS, Docker)

- Own autonomy software architecture, code reviews, and CI pipelines; lead 6 pathfinding developers and co-lead a 20-member software team through weekly scrums and cross-team coordination.
- Optimize route-planning algorithms under uncertain conditions to improve simulated sail-time efficiency.
- Build mock testing frameworks to validate autonomous navigation logic prior to hardware deployment.
- Coordinate autonomy design decisions with electrical and mechanical engineering leads.

Undergraduate Teaching Assistant

Sept 2025 – Present

UBC Department of Computer Science

- **CPSC 317: Computer Networking** (Jan 2026 – April 2026)
Lead networking tutorials (TCP/IP, Sockets) and troubleshoot systems-level C/Java environments.
- **CPSC 310: Software Engineering** (Sept 2024 – Dec 2025)
Mentored 50+ students on Node.js/TypeScript architecture; resolved complex Yarn and CI/CD issues.

Technical Projects

Reliable UDP | github.com/raghumanimehta/ReliableUDP

Jun 2025 – Present

Systems Project (C++, Linux/Unix Sockets)

- Engineering a reliable data transfer layer over UDP to replicate TCP guarantees (ACKs, sequence no.s).
- Developing custom sender/receiver components utilizing POSIX networking APIs for robust delivery.
- Handling edge cases including packet loss and out-of-order delivery to ensure reliable data delivery under packet loss and reordering.

Rusty File System | github.com/raghumanimehta/Rusty-File-System-

Dec 2025 – Present

Systems Project (Rust, Filesystems, OS Internals)

- Implement a user-space file system in Rust with custom inode, block, and metadata abstractions.
- Design core storage and directory management logic with an emphasis on correctness, safety, and clear separation of concerns.

OCR System | github.com/CPSC319-2025/Team-3-CCS

Jan 2025 – Apr 2025

Backend Project (Python, AWS, PostgreSQL)

- Designed a serverless backend pipeline using AWS Textract, Lambda, and SQS for parallel processing.
- Architected a high-availability system supporting 50+ concurrent users with zero data loss.